

Majorana Modes in the Machine

Project Summary — Kitaev Chain & Majorana Zero Modes on NISQ Hardware

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The Project in One Slide

The bridge: fermions $\rightarrow$ qubits $\rightarrow$ NISQ

- The **1D Kitaev chain** is the minimal model of a 1D topological superconductor: in the topological phase it hosts **Majorana zero modes** localized at its two ends.
- A **Jordan–Wigner** map turns that fermionic model into an exact qubit Hamiltonian — so a quantum device can, in principle, host the physics.
- We prepare its ground state by **VQE** and read the topological signature from a circuit-measurable observable.

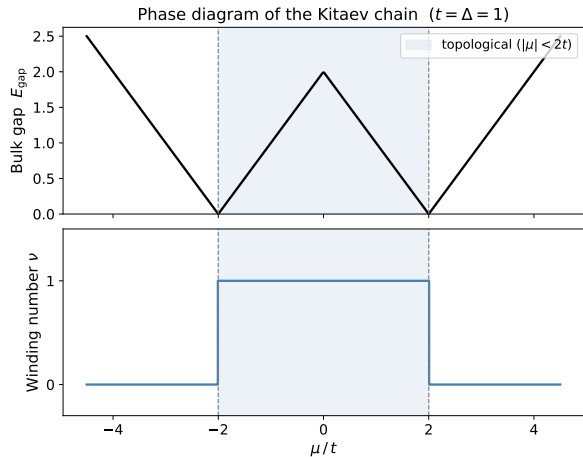
The question

Can a *topological signature* be prepared and still *survive* on a noisy NISQ device — and how far in system size L can it be pushed?

Four Blocks:

- 1 the physics bridge,
- 2 finite-size physics & qubit encoding,
- 3 measuring topology with circuits,
- 4 the NISQ reality check (results).

Block 1 — The Physics Bridge



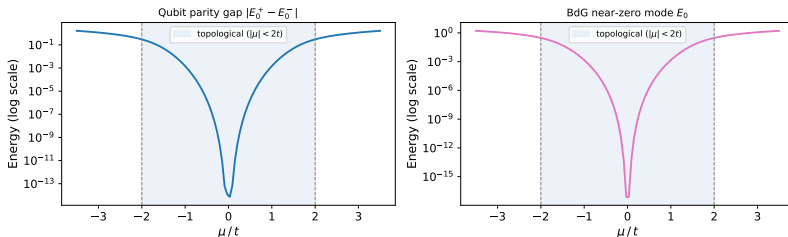
The Kitaev chain: p -wave pairing on a 1D lattice,

$$E_k = \sqrt{(\mu + 2t \cos k)^2 + (2\Delta \sin k)^2}.$$

- **Bulk gap** closes only at $|\mu| = 2t$ — the phase boundary.
- **Topological phase** for $|\mu| < 2t$: a $\mathbb{Z}_2$ **winding number** $\nu = 1$; trivial ($\nu = 0$) outside.
- Bulk–boundary correspondence $\Rightarrow$ **Majorana zero modes** pinned to the ends in the topological phase.

Block 2 — Finite-Size Physics & Qubit Encoding

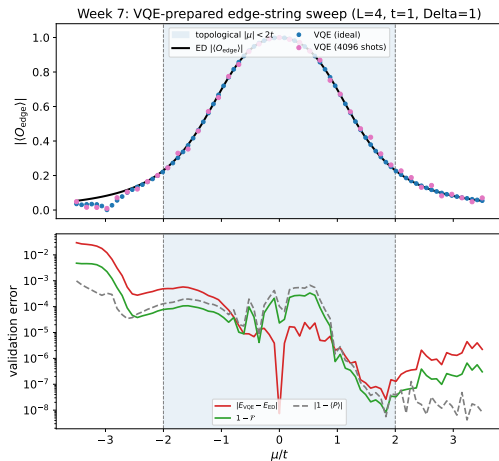
Qubit parity gap vs BdG splitting ($L = 10$, $t = \Delta = 1$, OBC)



On a finite chain the two end Majoranas overlap and split, $\Delta_P(L) \sim e^{-L/\xi}$.

- **Exponential protection:** the parity gap shrinks as the chain grows — longer is better.
- **Jordan–Wigner** gives the *exact* qubit Hamiltonian
$$H = -\frac{\mu}{2} \sum_j Z_j + \frac{t-\Delta}{2} \sum_j X_j X_{j+1} + \frac{t+\Delta}{2} \sum_j Y_j Y_{j+1}.$$
- **Parity** $\hat{P} = \prod_j Z_j$ is a $\mathbb{Z}_2$ symmetry: even/odd sectors label the two near-degenerate ground states.

Block 3 — Measuring Topology with Quantum Circuits (VQE)

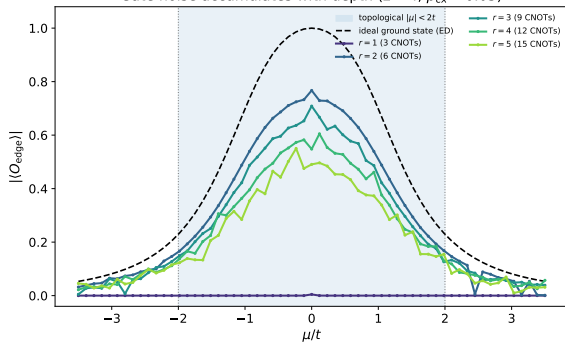


Prepare the ground state *variationally*, then detect the phase from a *circuit-measurable* observable.

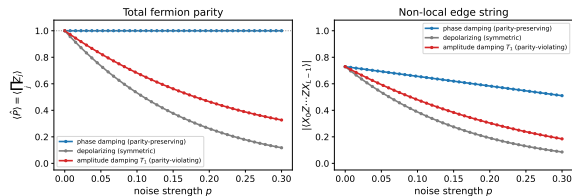
- **Parity-penalized VQE** (EfficientSU2) prepares the even-sector ground state — no exact eigenvector needed.
- Topology lives in a **non-local edge string** $O_{\text{edge}} = X_0 Z_1 \cdots Z_{L-2} X_{L-1}$.
- Sweeping μ : the string **turns on** in the topological phase, off in the trivial — it tracks the transition.

Block 4 — NISQ Reality Check I: Noise Breaks the Diagnostic

Gate noise accumulates with depth ($L = 4$, $p_{CX} = 0.05$)



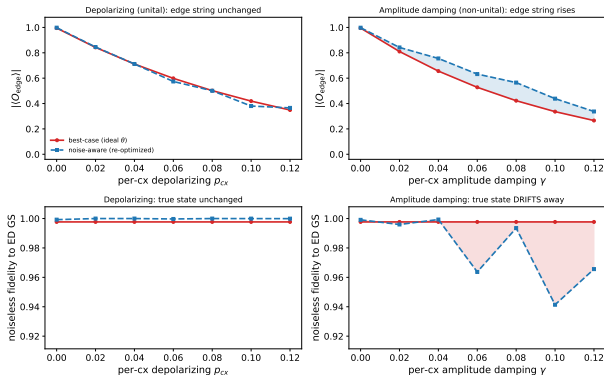
Parity (symmetry) survives dephasing; the edge string (topology) does not ($L = 6$, $\mu = 1t$)



Circuit-level gate noise enters *through the gates*: each cx adds an error channel, so the signal decays as $(1 - p_{CX})^{r(L-1)}$ with depth. **Left:** an under-expressive ansatz is dead everywhere; once expressive, deeper circuits are pushed down by accumulated CNOT noise. **Right:** **parity is a symmetry** (protected under dephasing), but the **edge string (topology) is not noise-immune** — it decays under every channel.

Block 4 — NISQ Reality Check II: Optimizing $\neq$ Preparing

Optimizing the noisy cost is not preparing the state: non-unital noise inflates the edge string while the true-state fidelity falls ($L = 4$, $r = 3$, $\mu = 0$, $\lambda = 0.1$)



With a NoiseModel-backed estimator, every VQE cost evaluation is corrupted.

- **The two rows:** *top* = measured edge string; *bottom* = **true-state fidelity** to the exact ground state.
- Non-unital noise (T_1 , *right*) **inflates the edge string** while **fidelity collapses** — the observable lies.
- Optimizing the noisy cost $\neq$ preparing the state.

Block 4 — NISQ Reality Check III: Scaling to Large L (Methods)

Past the dense $L \approx 15$ wall: remove every exponential object but the mild 2^L statevector.

Removing the two exponentials

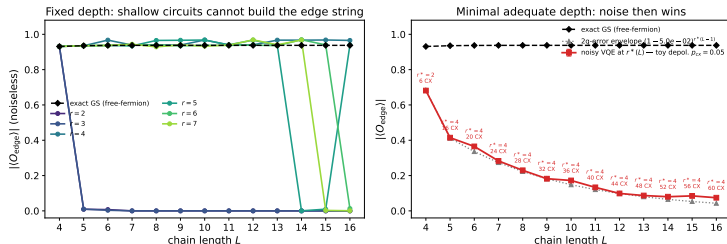
- **ED** $\rightarrow$ **free fermions**: the model is quadratic — exact GS & edge string from a $2L \times 2L$ **Bogoliubov–de Gennes** matrix, $O(L^3)$.
- 4^L **density matrix** $\rightarrow$ **MPS trajectories**: each noisy shot is one **matrix-product state (MPS)** trajectory — a chain of per-site $\chi \times \chi$ tensors with bond dimension $\chi \leq 2^r$ ($= 16$ at $r^*=4$), so memory is $O(L\chi^2)$: **linear in L** , not 4^L .

The mild object + HPC

- **Ideal VQE**: a 2^L statevector with *sparse* Pauli operators — never the 4^L dense H ($\sim 500\times$ faster at $L = 14$).
- $\mu = 0.5t$ (gapped), L-BFGS-B, ED-free even-sector selection.
- (L, reps) grid parallelized on the **Leipzig SC cluster**.

Block 4 — NISQ Reality Check IV: The Scaling Result

Two ways a fixed-depth preparation fails as L grows ($\mu = 0.5t$, noise: toy depol. $p_{CX} = 0.05$)

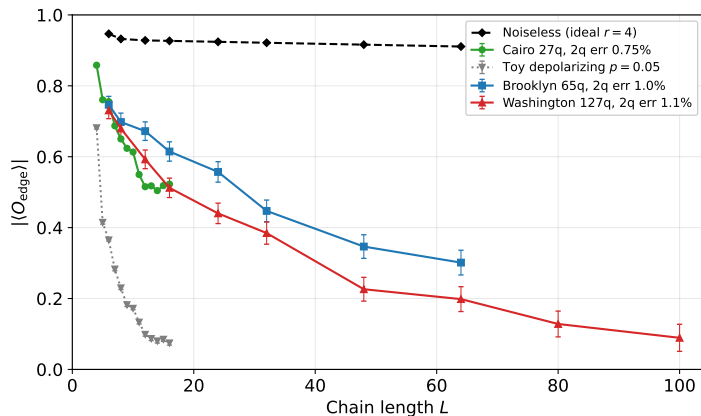


The two failure modes decouple.

- **Left:** expressibility is *cheap and L -independent* — a **constant** depth $r^* = 4$ reproduces the edge string to $L = 16$ (shallow $r = 2, 3$ fail).
- **Right:** at fixed depth $n_{\text{cnot}} = 4(L - 1)$ grows *linearly*, so **noise is the sole large- L wall** — the edge string decays exponentially.

Toy ($p = 0.05$): dead by $L \approx 14$. Real Cairo ($2q$ err ≈ 0.0075 , $\sim 6\times$ gentler) keeps $\langle O_{\text{edge}} \rangle$ *plateauing* ~ 0.5 to $L = 16$ — it **survives far further** (next slide).

Block 4 — NISQ Reality Check V: Real Hardware to $L = 100$



Every run at $r^* = 4$, one axis.

- **Noiseless** (black): flat ~ 0.9 — topological at every L .
- **Toy** $p=0.05$: $\sim 6\times$ too harsh, dead by $L = 16$.
- **Real IBM** (Cairo / Brooklyn / Washington): survive far further; **agree on the overlap**, ordered by 2q error.
- **Gone** (~ 0.09) only by $L = 100$ — decoherence over a *linear* CNOT count.

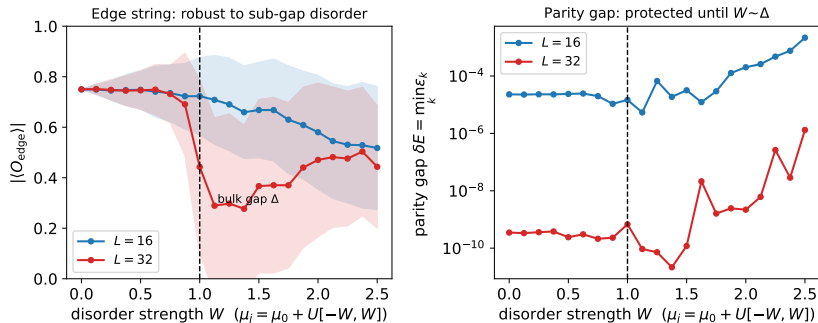
Block 4 — Which Physical Errors Kill the Topology?

A perturbation is harmless only if it is **(i) local**, **(ii) fermion-parity preserving**, and **(iii) weaker than the bulk gap Δ** . Break any one and the phase dies.

Physical error (Kitaev H)	Jordan–Wigner $\rightarrow$ qubit	Verdict
Charge noise $\delta\mu_i c_i^\dagger c_i$	$\sum_i \delta h_i Z_i$ (parity-even)	robust (up to Δ)
Quasiparticle poisoning $c_i^\dagger$	$(\prod_{k<i} Z_k)(X_i \mp iY_i)$ (parity-odd)	fatal : flips the bit
Gap closure $\mu \rightarrow 2t$	$\sum_i Z_i$ dominates $\sum_i X_i X_{i+1}$	fatal : phase transition

What to measure: the string order O_{edge} , the fermion parity $P = \prod_j Z_j$, and the splitting $\delta E = |E_{\text{even}} - E_{\text{odd}}| \sim e^{-L/\xi}$.

Block 4 — Robust Error: Local Disorder (Exact, to $L = 32$)

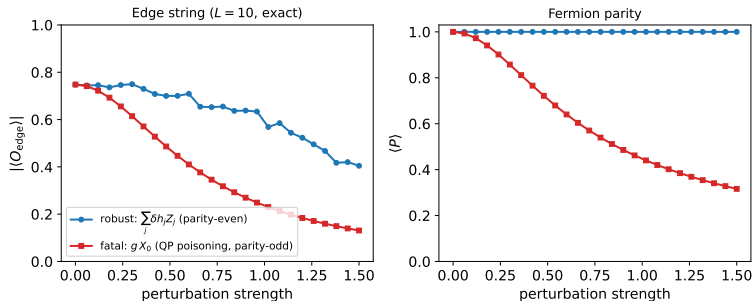


Charge noise = a random field $\sum_j \delta h_j Z_j$ (parity-even): per-site $\mu_j = \mu_0 + \text{Uniform}(-W, W)$, where W is the disorder half-width (the x-axis). Free-fermion BdG, exact.

- Below the bulk gap Δ the edge string **holds** and the splitting stays *exponentially* small.
- Past $W \sim \Delta$ the gap closes and both die.

Local, parity-even, sub-gap $\Rightarrow$ topologically harmless.

Block 4 — Fatal Error: Quasiparticle Poisoning

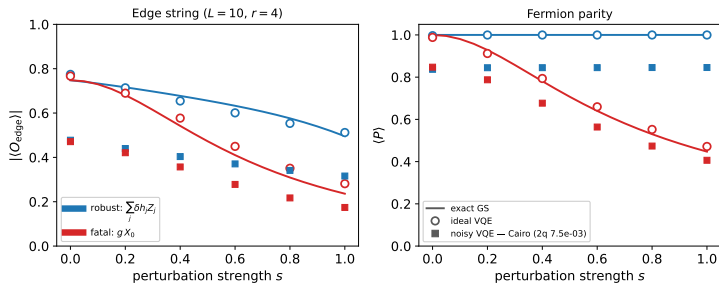


A stray electron $c_0 + c_0^\dagger = \gamma_0 = X_0$ is **parity-odd**. Exact diagonalisation, $L = 10$.

- Parity-even Z-disorder: $\langle P \rangle = +1$, edge string survives.
- Parity-odd $g X_0$: $\langle P \rangle$ and O_{edge} **collapse**.

QP poisoning is the Achilles heel: the poisoning time must beat the gate time.

Block 4 — The Physical Error on the Real Circuit (VQE)



The taxonomy on the *actual* VQE circuit ($L=10, r=4$): ideal statevector ($\circ$) and noisy Cairo density matrix ($\blacksquare$) over the exact curves.

- Ideal VQE **tracks the exact ground truth** for both channels — the qubit ansatz reproduces the physics.
- **Robust** Z-disorder: $\langle P \rangle$ stays +1; **fatal** $g X_0$ **collapses** $\langle P \rangle$ — the parity breakdown *on the qubits*.
- Cairo noise stacks *below* the ideal markers: physical error $\times$ gate error.

Takeaways

- The Kitaev chain's Majorana edge signature can be **prepared and measured on qubits** via a non-local string order $O_{\text{edge}} = X_0 Z_1 \cdots Z_{L-2} X_{L-1}$.
- **Parity (symmetry) is protected** under noise; the **edge string (topology) is not** noise-immune at fixed L .
- Classically, both exponentials are removable: the ED reference $\rightarrow$ **free fermions** (BdG), the 4^L density matrix $\rightarrow$ **MPS trajectories** — only a mild 2^L statevector remains.
- For the edge-string diagnostic, **expressibility is L -independent** (constant depth $r^* = 4$): the large- L wall is *decoherence* over a linearly growing gate count.
- On **real IBM noise** the signature survives far beyond the toy model ($p = 0.05$), staying measurable to $L \sim 48$ across devices.
- **Which errors kill topology?** Local parity-even sub-gap noise is harmless; **quasiparticle poisoning** (parity-odd) and gap closure are fatal.

Notes: `scaling_to_large_L.tex`, `kitaev_error_taxonomy.tex`,
`parity_topology_protection.tex`.